Monotone Convergence Theorem

An End-to-End Meta-Math Walkthrough

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(Dated: October 2026)

Abstract

Test the limit against one simple function at a time. A factor $c < 1$ makes the eventual comparison with f_n valid even where the limit equals the test function. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

FA-01 / Functional Analysis and Measure Theory / Report v1

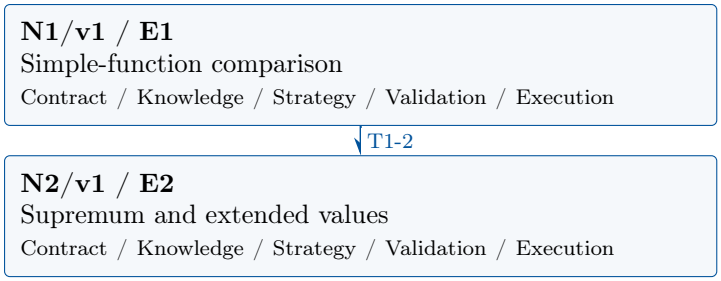
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *On any measure space (X, Σ, μ), let $f_n : X \to [0, \infty]$ be measurable with $f_n \le f_{n+1}$ everywhere. For $f = \sup_n f_n$, prove $\int f \, d\mu = \lim_n \int f_n \, d\mu$, allowing $+\infty$. No finiteness or sigma-finiteness of μ is assumed.*

A. Foundations and permitted imports

Use countable additivity, continuity from below for measures, and the definition $\int h = \sup\{\int s : 0 \le s \le h,\ s \text{ finite-valued measurable simple}\}$. For disjoint measurable A_i and $a_i \ge 0$, $\int \sum_i a_i 1_{A_i} = \sum_i a_i \mu(A_i)$, with $0 \cdot \infty = 0$. Continuity from below follows by decomposing an increasing union into disjoint successive differences. Integral monotonicity follows directly from the supremum definition.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1-2: N1/E1/v1 to N2. For every such s and c, $\lim_n \int f_n \geq c \int s$.

Receiving activity. N2.1

Compatibility. E1/v1 and $0 \leq f_n \leq f$; the integral is defined by simple minorants. Check: Never subtract two infinite integrals; handle the extended-real comparison directly.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Simple-function comparison

Object and role. Approximate the support of each positive level from below by places where f_n has reached cs. **Local obligation:** For every such s and c, $\lim_n \int f_n \geq c \int s$.

Inputs. A fixed simple $s = \sum_{i=1}^{m} a_i 1_{A_i} \leq f$ with disjoint A_i, and $0 < c < 1$.

C — Contract

Publish E1/v1 only after establishing this obligation: For every such s and c, $\lim_n \int f_n \geq c \int s$.

K — Knowledge

A fixed simple $s = \sum_{i=1}^{m} a_i 1_{A_i} \leq f$ with disjoint A_i, and $0 < c < 1$. Retain the stated assumptions and import versions.

S — Strategy

Approximate the support of each positive level from below by places where f_n has reached cs.

V — Validation

At a point with $s > 0$, the strict margin $cs < s \leq f$ is essential; do not claim eventual attainment of s itself.

E — Execution

Carry out activities N1.1–N1.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N1.1 The supremum of measurable functions is measurable, so f is measurable. Set $B_n = \{x : f_n(x) \geq cs(x)\}$. These measurable sets increase. Every point with $s(x) > 0$ lies in their union, since otherwise $f(x) \leq cs(x) < s(x)$; points with $s(x) = 0$ already lie in every B_n.

N1.2 Because $cs1_{B_n} \leq f_n$, monotonicity gives $\int f_n \geq c \sum_i a_i \mu(A_i \cap B_n)$. Continuity from below gives $\mu(A_i \cap B_n) \uparrow \mu(A_i)$. The sum has finitely many nonnegative terms, so its limit is $\int s$, also when a positive term is infinite. This proves the export.

Export E1/v1. For every such s and c, $\lim_n \int f_n \geq c \int s$.

Acceptance gate. At a point with $s > 0$, the strict margin $cs < s \leq f$ is essential; do not claim eventual attainment of s itself.

Targeted repair. If a candidate uses $c = 1$, restore $0 < c < 1$ and repeat the support exhaustion before taking any limit. Withhold this export and recheck its descendants after a versioned correction.

B. N2/v1: Supremum and extended values

Object and role. Take the scalar limit $c \uparrow 1$, then the supremum over simple functions. **Local obligation:** $\lim_n \int f_n = \int f$ in $[0, \infty]$.

Inputs. E1/v1 and $0 \le f_n \le f$; the integral is defined by simple minorants. Incoming proof exports: E1/v1 at N2.1.

C — Contract

Publish E2/v1 only after establishing this obligation: $\lim_n \int f_n = \int f$ in $[0, \infty]$.

K — Knowledge

E1/v1 and $0 \le f_n \le f$; the integral is defined by simple minorants. Incoming proof exports: E1/v1 at N2.1. Retain the stated assumptions and import versions.

S — Strategy

Take the scalar limit $c \uparrow 1$, then the supremum over simple functions.

V — Validation

Never subtract two infinite integrals; handle the extended-real comparison directly.

E — Execution

Carry out activities N2.1–N2.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N2.1 Put $L = \lim_n \int f_n$, which exists in $[0, \infty]$ because the integrals increase. E1 gives $L \ge c \int s$. If $\int s < \infty$, let $c \uparrow 1$ to obtain $L \ge \int s$. If $\int s = \infty$, any fixed $c > 0$ already gives $L = \infty$.

N2.2 Taking the supremum over all simple $s \le f$ yields $L \ge \int f$. Conversely $f_n \le f$ gives $\int f_n \le \int f$ for every n, so $L \le \int f$. The two comparisons prove the asserted equality without an infinity subtraction.

Export E2/v1. $\lim_n \int f_n = \int f$ in $[0, \infty]$.

Acceptance gate. Never subtract two infinite integrals; handle the extended-real comparison directly.

Targeted repair. If infinity is treated as an ordinary real number, replace that manipulation with the two order inequalities just proved. Withhold this export and recheck its descendants after a versioned correction.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — Simple-function comparison

1. **N1.1.** The supremum of measurable functions is measurable, so f is measurable. Set $B_n = \{x : f_n(x) \geq cs(x)\}$. These measurable sets increase. Every point with $s(x) > 0$ lies in their union, since otherwise $f(x) \leq cs(x) < s(x)$; points with $s(x) = 0$ already lie in every B_n.

2. **N1.2.** Because $cs1_{B_n} \leq f_n$, monotonicity gives $\int f_n \geq c \sum_i a_i \mu(A_i \cap B_n)$. Continuity from below gives $\mu(A_i \cap B_n) \uparrow \mu(A_i)$. The sum has finitely many nonnegative terms, so its limit is $\int s$, also when a positive term is infinite. This proves the export.

B. N2 — Supremum and extended values

1. **N2.1.** Put $L = \lim_n \int f_n$, which exists in $[0, \infty]$ because the integrals increase. E1 gives $L \geq c \int s$. If $\int s < \infty$, let $c \uparrow 1$ to obtain $L \geq \int s$. If $\int s = \infty$, any fixed $c > 0$ already gives $L = \infty$.

2. **N2.2.** Taking the supremum over all simple $s \leq f$ yields $L \geq \int f$. Conversely $f_n \leq f$ gives $\int f_n \leq \int f$ for every n, so $L \leq \int f$. The two comparisons prove the asserted equality without an infinity subtraction.

□

V. WORKED ORCHESTRATION

For Lebesgue measure on $(0, 1)$, take $f_n(x) = \min(n, 1/x)$. It increases to $1/x$. Splitting at $1/n$ gives $\int_0^1 f_n = 1 + \log n$ for $n \geq 1$, hence the theorem correctly returns $\int_0^1 dx/x = \infty$. This exercises the extended-valued conclusion.

A. Boundary and hypothesis audit

On $(0, 1)$, $g_n = n1_{(0,1/n)}$ tends pointwise to zero but has integral one. The sequence is not increasing; pointwise convergence alone cannot replace the monotonicity hypothesis.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Rejected argument, diagnosis, and correction

Rejected candidate (N1.1): $s \le f$ implies eventually $s \le f_n$. Take $s = f = 1$ and $f_n = 1 - 1/(n+1)$. **Diagnosis:** Strategy lost the strict comparison margin. **Repair:** compare cs for $c < 1$, then use N2 to pass $c \uparrow 1$. This fills the proof under the original hypotheses. Recheck E1 and E2; the target is unchanged.

B. Publication and reuse

The corrected export retains its explicit assumptions. A hypothetical revision receives a new version, and each consuming activity is rechecked; this scenario is not a recorded runtime event. **Next direction:** Use this export to prove Fatou's lemma by applying it to the increasing tail infima of a nonnegative sequence.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. MIT OpenCourseWare, Introduction to Functional Analysis: lecture notes. See the lectures on integration, Hilbert spaces, Baire category, and Hahn–Banach.